Searching Legislative Design Space for Compromise and Productivity

A Computational Framework for Comparing Congress-Style Institutional Designs

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Legislatures need enough productivity to act, but productivity alone can also mean volatility, agenda capture, or weakly supported lawmaking. This paper presents a reproducible computational framework for searching legislative design space rather than defending any single reform. The simulator routes shared synthetic legislatures and bill streams through Congress-style institutional bundles that vary agenda access, voting thresholds, alternative selection, citizen review, lobbying safeguards, affected-group protection, package bargaining, and reversibility.

The contribution is methodological. The simulator separates productivity from compromise, public alignment, weak public-mandate passage, policy yield per submitted proposal, pre-floor blockage reliance, proposer advantage, concentrated harm, lobbying capture, administrative cost, and correction over time. One main comparison campaign generates the central scenario tables and scatter figures, while separate scripted diagnostics generate seed, ablation, manipulation-stress, chamber-structure, and empirical-bridge checks. The current experiment is breadth-first: default enactment remains one burden-shifting stress test, while the main comparison spans conventional, deliberative, anti-capture, agenda-scarcity, policy-tournament, harm-protection, and review-oriented families. The paper argues for mechanism search: legislative designs should be compared as portfolios of incentives and safeguards, not as isolated passage thresholds.

CCS Concepts: • **Human-centered computing → Collaborative and social computing theory, concepts and paradigms**; • **Computing methodologies → Modeling and simulation**; • **Applied computing** → *Law*.

Additional Key Words and Phrases: legislative institutions, institutional design, computational simulation, agenda control, compromise, collective decision-making, voting systems

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1 Introduction

The motivating question is not whether one unusual voting rule should replace Congress. It is whether systematic design search can help identify legislative structures that better combine productivity, compromise, and representative responsiveness. A legislature can be unified without being democratic; authoritarian systems can appear cohesive because disagreement is suppressed. A useful representative system should instead help actors with competing interests reach compromises connected to public support, public benefit, and the burdens imposed on affected groups.

Default enactment, where a proposal passes unless a blocking coalition stops it, is one design in the search space, not the object of the project. It is useful as a stress test because it reverses the usual burden of coalition formation. The broader object is the mechanism bundle around any final yes-or-no vote: who may propose, how agenda access is rationed, whether information is improved before voting, whether opponents have scarce challenge rights, whether bill content can move toward compromise, whether concentrated losers receive protection, and whether bad laws can be corrected later.

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This paper therefore makes a modest claim. It does not validate a new constitution or forecast the U.S. Congress. It contributes an executable framework for comparing legislative mechanisms under common assumptions, with explicit metrics and reproducible artifacts. The simulations are comparative stress tests: they reveal when a rule produces output, when it produces compromise, and when those two diverge.

2 Related Work

The model draws on individual-level simulation, spatial voting, and institutional theory. Agent-based and computational political models are useful when heterogeneous actors interact under rules that create aggregate outcomes [de Marchi and Page 2014; Epstein 2006; Gilbert and Troitzsch 2005]. Spatial voting motivates one-dimensional ideal points and status-quo-relative utility [Downs 1957]. Legislative bargaining and agenda-setter models motivate explicit proposer identity and proposer gain [Baron and Ferejohn 1989; Romer and Rosenthal 1978]. Pivotal-politics and veto-player theories motivate comparing simple majority, supermajority, bicameral, and executive-veto systems [Krehbiel 1998; Tsebelis 2002]. Committee and agenda-control theory motivates treating floor denial, referral, information, and amendment control as outcomes rather than missing observations [Cox and McCubbins 2005; Gilligan and Krehbiel 1987; Krehbiel 1991; Shepsle 1979; Shepsle and Weingast 1987; Weingast and Marshall 1988].

Several tested mechanisms come from adjacent literatures. Lobbying is modeled as access, effort, information, public pressure, and participation bias rather than only direct vote buying [Austen-Smith 1993, 1995; de Figueiredo and Richter 2014; Gilens and Page 2014; Hall and Deardorff 2006; Hall and Wayman 1990]. Citizen-panel variants draw on deliberative polling and mini-public practice [Fishkin 2009; Organisation for Economic Co-operation and Development 2020]. Agenda-credit variants echo quadratic-voting and storable-vote work on expressing intensity under scarcity [Casella 2005; Lalley and Weyl 2018]. Reversibility variants draw on temporary legislation and sunset review [Gersen 2007; Ranchordas 2014]. Policy tournaments draw on Condorcet and agenda-manipulation problems [McKelvey 1976; Tideman 1987]. Model documentation follows ODD and ODD+D conventions [Grimm et al. 2010; Müller et al. 2013].

Default-effect rules have real analogues, although usually in bounded domains: WTO dispute settlement, EU reverse-qualified-majority enforcement, tacit-acceptance procedures, and U.S. fast-track or disapproval structures all shift procedural burdens in limited ways [Churchill and Ulfstein 2000; Davis 2024; European Parliament and Council of the European Union 2011; United States Congress 1996; World Trade Organization 1994]. These analogues suggest the same design lesson as the simulator: burden-shifting rules need guardrails around proposal access, objection, information, and correction.

3 Simulator

Each run creates a synthetic legislature, bill stream, status quo, and institutional process. Legislators have ideal points in $[-1, 1]$, party labels, district preferences, and heterogeneous sensitivity to party pressure, constituency pressure, lobbying, reputation, and compromise. Bills have a proposer, policy position, public support, generated public benefit, lobby pressure, salience, private gain, issue domain, public-benefit uncertainty, affected-group support, and concentrated harm. Enacted bills update the status quo, so votes evaluate policy movement rather than isolated proposals.

The implementation separates voting rules from legislative processes. Voting rules interpret yes/no totals. Processes wrap a bill in stages such as proposal access, leadership control, committee assignment and gatekeeping, discharge-petition bypass, open floor calendars, citizen petitions, bicameralism, conference, veto, judicial review, coalition confidence, initiatives, lotteries, attention budgets, challenge vouchers, routing, norm erosion, sunset review, credits, bonds, lobbying, harm review, package bargaining, mediation, alternatives, amendment tournaments, citizen-panel review, objection, and law-registry review. Post-enactment review now tracks implementation delay, capacity, underfunding, nonenforcement, and renewal pressure as separate diagnostics. This modularity is central: the same threshold behaves differently when proposals are open, filtered, challenged, revised, reviewed, or made reversible.

The stylized U.S.-like conventional benchmark uses House majority passage, a Senate 60 percent threshold, presidential veto, two-thirds override, and an agenda-access gate. The gate prevents the benchmark from treating every introduced bill as a floor bill; it schedules bills using observable proxies: public support, salience, policy stability, sponsor-party fit, partisan queue conflict, capture risk, concentrated harm, and uncertainty. It does not observe generated public benefit directly. This makes the conventional baseline pass the current calibration screen without claiming it is a fitted model of Congress.

4 Metrics and Campaign

The simulator reports a dashboard rather than a single welfare score. Higher-is-better metrics include productivity, enacted support, generated welfare among enacted bills, policy yield, compromise, public alignment, legitimacy, citizen certification, district alignment, and anti-lobbying success. Lower-is-better metrics include gridlock, low-support or weak-mandate passage, popular failure, lobbying capture, public-preference distortion, minority harm, proposer gain, administrative cost, blockage reliance, low-support active-law share, and false-negative default passage. Floor consideration, access denial, committee rejection, challenge, amendment, compensation, reversal, attention spending, objection, and policy shift are diagnostic: high values can mean useful activity or institutional strain.

The metric types matter. Some metrics are structural outputs, such as productivity, enactment counts, floor load, review load, pre-floor blockage reliance, and administrative cost. Some are synthetic value proxies generated inside the model, such as public support, public benefit, legitimacy, weak public mandate, policy yield, and affected-group harm. Others are diagnostics, such as amendment overload, poison-pill risk, petition distortion, implementation failure, challenge, objection, and reversal rates. Chamber low-support passage is mainly a threshold-failure metric; under ordinary majority voting it is often mechanically low. Weak public-mandate passage is the cross-system public-support failure metric because it asks whether enacted bills had generated public support below 50 percent, regardless of the chamber vote rule.

For display, the paper reports a directional score only as a diagnostic reading aid. Representative quality now combines enacted-policy quality, policy yield per submitted proposal, and public-mandate legitimacy, so a system that looks good mainly by blocking most proposals is distinguished from one that actually produces public-value output. Risk control inverts chamber low-support passage, weak public-mandate passage, minority harm, lobbying capture, public-preference distortion, concentrated-harm passage, proposer gain, and policy shift. Administrative feasibility inverts estimated procedural load from attention spending, review, challenge, amendment, and alternative selection. The directional score averages productivity, representative quality, risk control, and administrative feasibility, but the analysis emphasizes Pareto tradeoffs because different value profiles favor different institutional designs.

All main results come from one reported comparison campaign. It combines broad assumptions, adversarial generator cases, weighted party-system cases, and a stylized rising-contention timeline in one CSV. Cases vary polarization, party loyalty, compromise culture, lobbying, constituency pressure, party count, proposal pressure, capture pressure, anti-lobbying reform demand, and six adversarial bill streams: high-benefit extreme reform, popular harmful bills, moderate capture, delayed-benefit reform, concentrated rights-like harm, and anti-lobbying backlash.

Table 1. Compact design-space index for the main comparison campaign. Labels match Table 3 and the scatter plots; detailed mechanism descriptions are in the appendix and supplement.

Family	Labels	Mechanism theme	Primary stress risk
Conventional	SM S60 BIC VETO CUR	Thresholds/vetoes	Bottlenecks
Procedure	LEAD PROC COMM DIS	Agenda control	Queue capture
Courts	COURT	Review architecture	Juridification
Public model	DIST	District publics	Proxy opinion
Agenda gates	SCR LOT OPEN	Screen/lottery/calendar	Gate bias
Committee design	PCOM	Proportional assignment	Committee capture
Coalition	PARL	Confidence/access	Bloc cartels
Direct democracy	INIT PET	Initiative/petition paths	Campaign bias
Tournaments	PAIR AMT	Alternatives/amendments	Clones/decoys
Mini-public	JURY RPAN	Citizen burden shift	Manipulation
Scarcity	QAB BOND	Credits/bonds	Hoarding
Harm rules	HARM COMP	Minority protection	Holdouts
Bargaining	PKG MPKG OMNI	Package trades	Proxy limits
Correction	LAW OBJ	Review/objection	Instability
Anti-capture	CAP ACG INFL	Lobby safeguards	Evasion
Adaptive	RISK NORM LEARN	Risk/norm/learning	Gaming
Portfolio	PORT XPORT	Mechanism bundle	Complexity
Default stress	DP DPC DPM	Burden shift	False silence
Out of scope	—	Elections/media/etc.	Endogeneity

Scenario selection follows a fixed coverage rule: include conventional baselines, the stylized U.S.-like benchmark, readable representatives for implemented non-default design classes, and three default-enactment stress tests. The selection manifest records each row's role and rationale; `make family-screen` reports fixed-rule family champions. The main comparison now includes pre-floor mechanisms directly: proportional committees, discharge petitions, open floor calendars, citizen petitions, amendment tournaments, and anti-capture proposal access. Earlier default-pass sweeps remain side logs, not the organizing evidence base.

The chamber-structure supplement tests representation architecture, committee power, eligibility filters, and ex ante review. It is separate because it asks how those structures shift the same productivity/compromise tradeoff.

Calibration is a screening step, not a parameter fit. `make calibrate` compares conventional baselines with broad benchmark ranges derived from Voteview, Congress.gov, govinfo, Comparative Agendas Project, ParlGov, lobbying-disclosure and OpenFEC records, Center for Effective Lawmaking measures, and V-Dem indicators [Comparative Agendas Project 2026; Döring and Manow 2024; Federal Election Commission 2026; Lewis et al. 2021; Library of Congress 2026; United States Government Publishing Office 2020; United States Senate Office of Public Records 2026; V-Dem Institute 2026; Volden and Wiseman 2014]. These are proxy checks for gross plausibility. A separate empirical bridge now reads a documented 180-row Congress.gov bill-progression sample for floor load, committee reporting, and enactment rate; other raw datasets remain absent. The current calibration screen passes all tracked checks, including the stylized U.S.-like benchmark's enactment rate and floor load, but the bridge should be read as an early validation foothold rather than model fitting.

5 Results

Table 3 compares representative family examples across broad and adversarial synthetic cases; the appendix reports the full table with directional and seed diagnostics. Table 4 shows the narrow productivity/compromise/risk frontier. It is

Table 2. Chamber, apportionment, committee, and review family champions from the chamber-structure supplement. Values are averaged across the chamber campaign cases.

Label	Family champion	Prod.	Comp.	Risk	Malapp.
BICAM	Bicameral conflict	0.31	0.27	0.86	0.00
ASSIGN	Committee assignment	0.22	0.29	0.89	0.00
COMMP	Committee power	0.25	0.28	0.87	0.00
CONV	Conventional benchmark	0.34	0.25	0.85	0.00
REVIEW	Independent review	0.34	0.25	0.85	0.00
ROUTE	Origination and routing	0.31	0.27	0.86	0.00
OTHER	Other chamber design	0.31	0.26	0.86	0.00
SEAT	Seat allocation	0.34	0.25	0.85	0.00
SELECT	Selection and retention	0.36	0.24	0.84	0.00
UPPER	Upper-chamber composition	0.36	0.26	0.84	0.12

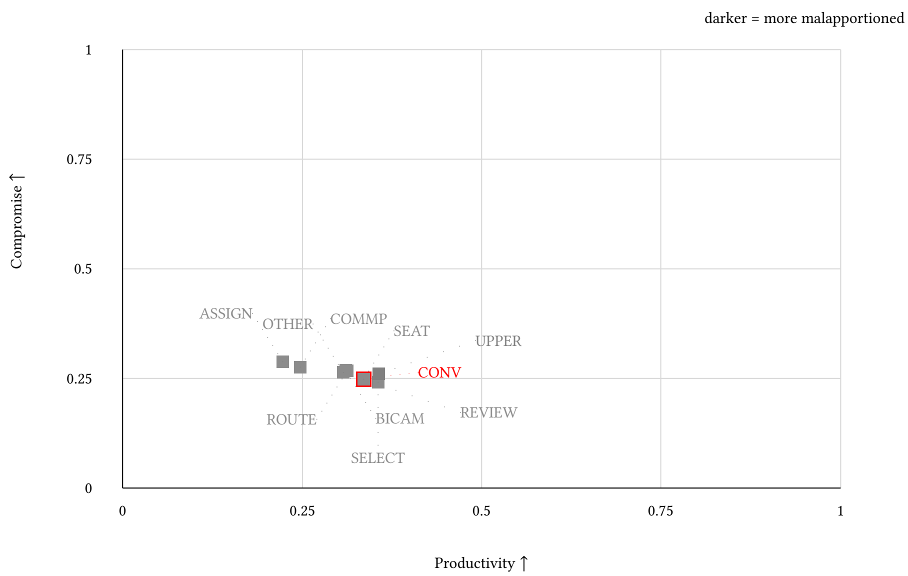


Fig. 1. Chamber-structure family champions from the supplemental chamber campaign. The axes keep the paper's central comparison visible, while grayscale intensity marks the generated malapportionment index. The figure is a structural sensitivity view, not a replacement for the main scenario comparison.

not a recommendation because it excludes administrative feasibility and compresses risk into one objective. Table 5 adds ablation and manipulation-stress diagnostics, while Table 6 separates the limited empirical bridge from the synthetic mechanism findings.

The main results are tradeoffs, not rankings. Pairwise alternatives are the strongest non-default example: they raise productivity and compromise while reducing weak public-mandate passage, but stress diagnostics show procedural cost and clone/decoy vulnerability. Backward-to-floor systems sharpen the interpretation. Petition and open-calendar variants preserve access while routing risk to deliberation; discharge and proportional-committee variants expose committee-design effects; anti-capture access filters before floor consideration; and amendment tournaments test content control. The portfolio hybrid is a design hypothesis, not a winner. Open default enactment has the highest

Table 3. Representative family comparison from the main campaign. Entries are mean±case half-range; full names and seed diagnostics are in the appendix and supplement.

Label	Representative system	Prod. ↑	Yield ↑	Comp. ↑	Weak mand. ↓	Block dep. ↓	Risk ctrl. ↑	Admin ↓
CUR	**Stylized U.S.-like benchmark**	**0.052±0.103**	**0.033±0.064**	**0.368±0.184**	**0.007±0.022**	**0.518±0.138**	**0.929±0.045**	**0.016±0.014**
SM	Simple majority	0.320±0.458	0.173±0.149	0.245±0.193	0.213±0.291	0.000±0.000	0.852±0.087	0.070±0.000
COMM	Committee regular order	0.207±0.409	0.116±0.127	0.287±0.184	0.098±0.119	0.491±0.260	0.891±0.099	0.032±0.022
PCOM	Proportional committees	0.234±0.426	0.128±0.131	0.272±0.192	0.153±0.122	0.453±0.261	0.873±0.088	0.056±0.008
DIS	Discharge petition	0.137±0.168	0.095±0.118	0.314±0.178	0.041±0.087	0.553±0.119	0.925±0.049	0.075±0.028
PARL	Parliamentary coalition	0.225±0.425	0.133±0.161	0.292±0.267	0.068±0.196	0.482±0.285	0.904±0.114	0.019±0.030
INIT	Citizen initiative	0.430±0.385	0.237±0.106	0.222±0.209	0.225±0.280	0.000±0.000	0.834±0.073	0.070±0.000
PET	Citizen agenda petition	0.196±0.188	0.124±0.111	0.254±0.182	0.079±0.110	0.384±0.192	0.911±0.032	0.047±0.062
PAIR	Pairwise alternatives	0.542±0.206	0.335±0.144	0.502±0.093	0.002±0.002	0.291±0.114	0.937±0.067	0.285±0.044
AMT	Amendment tournament	0.537±0.205	0.333±0.142	0.502±0.093	0.002±0.003	0.295±0.115	0.937±0.067	0.284±0.044
JURY	Citizen jury gate	0.196±0.180	0.131±0.122	0.282±0.216	0.062±0.238	0.000±0.000	0.906±0.080	0.190±0.000
RPAN	Random public panel	0.219±0.205	0.141±0.118	0.272±0.213	0.077±0.156	0.000±0.000	0.898±0.067	0.190±0.000
OPEN	Open floor calendar	0.352±0.368	0.183±0.119	0.253±0.163	0.252±0.293	0.036±0.065	0.842±0.069	0.098±0.027
QAB	Attention budget	0.246±0.366	0.139±0.140	0.265±0.191	0.177±0.287	0.267±0.132	0.878±0.077	0.220±0.032
BOND	Proposal bond	0.319±0.459	0.173±0.151	0.246±0.195	0.211±0.287	0.006±0.013	0.852±0.087	0.070±0.002
HARM	Harm-weighted majority	0.271±0.449	0.153±0.147	0.262±0.194	0.202±0.294	0.000±0.000	0.876±0.070	0.070±0.000
MPKG	Package bargaining	0.343±0.446	0.179±0.142	0.241±0.199	0.220±0.198	0.000±0.000	0.833±0.079	0.173±0.116
CAP	Anti-capture bundle	0.307±0.187	0.198±0.145	0.247±0.177	0.152±0.223	0.047±0.147	0.884±0.057	0.065±0.016
ACG	Anti-capture access	0.271±0.218	0.168±0.143	0.250±0.180	0.186±0.256	0.082±0.229	0.885±0.059	0.061±0.024
RISK	Risk-routed majority	0.530±0.330	0.259±0.162	0.264±0.170	0.232±0.173	0.000±0.000	0.841±0.070	0.171±0.029
PORT	Portfolio hybrid	0.504±0.299	0.321±0.291	0.421±0.063	0.031±0.026	0.254±0.167	0.932±0.034	0.236±0.094
XPORT	Expanded portfolio	0.783±0.237	0.439±0.290	0.463±0.052	0.003±0.004	0.062±0.141	0.904±0.054	0.586±0.094
LAW	Law registry	0.348±0.466	0.186±0.158	0.236±0.175	0.247±0.311	0.000±0.000	0.833±0.091	0.109±0.047
DP	Default enactment	0.866±0.088	0.430±0.223	0.136±0.220	0.472±0.369	0.000±0.000	0.629±0.065	0.070±0.000
DPM	Default + mediation	0.802±0.170	0.364±0.205	0.243±0.172	0.374±0.225	0.000±0.000	0.723±0.064	0.206±0.042

Note: Yield is welfare per submitted proposal, so systems do not receive the same interpretation for merely blocking most bills. Block dep. is pre-floor blockage reliance. Case half-ranges are descriptive variation across campaign cases, not statistical confidence intervals. Red marks the stylized U.S.-like conventional benchmark.

Table 4. Pareto-front scenarios under productivity, compromise, and risk control.

Label	Role	Prod. ↑	Comp. ↑	Risk ↑	Weak mand. ↓	Admin ↓
DP	Throughput endpoint	0.866	0.136	0.629	0.472	0.070
DPM	Throughput with mediation/challenge	0.802	0.243	0.723	0.374	0.206
XPORT	Non-dominated profile	0.783	0.463	0.904	0.003	0.586
PAIR	Alternative-comparison endpoint	0.542	0.502	0.937	0.002	0.285
AMT	Non-dominated profile	0.537	0.502	0.937	0.002	0.284

Note: The frontier changes if administrative feasibility, welfare, or rights/harm priorities are hard objectives.

Table 5. Ablation and manipulation-stress diagnostics for selected mechanisms.

Type	Mechanism	Comparison	Dir.	Comp.	Weak mand.	Admin
Ablation	Policy tournament	No alternatives	+0.052	+0.262	+0.206	+0.215
Ablation	Risk routing	No jury lane	-0.002	+0.000	+0.001	+0.009
Ablation	Anti-capture	Screen only	+0.023	+0.000	+0.028	+0.005
Ablation	Package bargaining	Scalar package	-0.015	-0.002	+0.002	+0.063
Ablation	Reversibility	No registry	-0.008	-0.009	-0.035	+0.039
Stress	Tournament attack	Clones/decoys	+0.087	+0.046	+0.010	-0.077
Stress	Citizen-panel manipulation	Noisy panel	-0.002	+0.018	-0.005	+0.000
Stress	Bad-faith harm claims	Loose claims	-0.002	-0.018	-0.018	+0.000
Stress	Objection astroturf	Astroturf/noise	+0.012	-0.006	+0.003	+0.010
Stress	Agenda flooding	Proposal flood	+0.008	-0.011	+0.007	-0.001

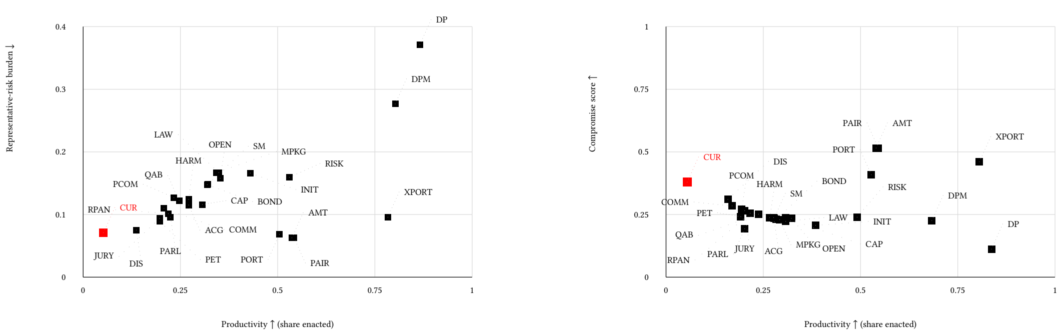
Note: Ablation weak mandate is a reduction, so positive is better; admin is added cost, so positive is worse. For stress rows, positives are losses or added burden.

Table 6. Empirical validation bridge. Raw inputs are optional; the table records whether each real-world signal has a local raw summary and the current simulator proxy used for calibration screening.

Signal	Raw	Simulator proxy	Status
Bill attrition	enactmentRate=0.017	current-system/productivity	ready
Floor consideration	floorLoad=0.122	current-system/floor	ready
Committee reporting from bill actions	committeeReportRate=0.083	current-system/floor	ready
Roll-call coalition size	coalitionSize=—	current-system/averageEnactedSupport	missing raw
Party unity	partyUnity=—	current-system/averageEnactedSupport	missing raw
Sponsor success concentration	sponsorSuccessGini=—	current-system/proposerAccessGini	missing raw

throughput but the largest weak-mandate burden. The stylized U.S.-like benchmark has low productivity and low weak-mandate passage; Table 3's yield and blockage-dependence columns distinguish gatekeeping success from policy production.

The chamber-structure supplement adds a flatter lesson: architecture changes the tradeoff surface, but no structural family dominates the mechanism-level campaign. Selection/retention filters and independent-review bundles score well; malapportioned upper chambers can raise productivity in some stressed cases, but the supplement exposes their representational cost.



(a) Representative-risk burden

(b) Compromise score

Fig. 2. Two productivity tradeoff views for the representative systems in Table 3, rendered side by side with the same 0–1 productivity axis. Panel (a) plots representative-risk burden, defined as 1−risk control; lower is better. Panel (b) plots compromise under weighted party-system sensitivity cases; higher is better. The stylized U.S.-like benchmark is red and marked with a larger square.

Label	Productivity	Compromise	Rep. quality	Risk ctrl.	Admin feas.
CUR	**0.05**	**0.37**	**0.48**	**0.93**	**0.98**
SM	0.32	0.25	0.42	0.85	0.93
COMM	0.21	0.29	0.44	0.89	0.97
PCOM	0.23	0.27	0.43	0.87	0.94
DIS	0.14	0.31	0.46	0.93	0.93
PARL	0.22	0.29	0.43	0.90	0.98
INIT	0.43	0.22	0.43	0.83	0.93
PET	0.20	0.25	0.44	0.91	0.95
PAIR	0.54	0.50	0.48	0.94	0.71
AMT	0.54	0.50	0.48	0.94	0.72
JURY	0.20	0.28	0.45	0.91	0.81
RPAN	0.22	0.27	0.45	0.90	0.81
OPEN	0.35	0.25	0.41	0.84	0.90
QAB	0.25	0.26	0.43	0.88	0.78
BOND	0.32	0.25	0.42	0.85	0.93
HARM	0.27	0.26	0.42	0.88	0.93
MPKG	0.34	0.24	0.41	0.83	0.83
CAP	0.31	0.25	0.43	0.88	0.93
ACG	0.27	0.25	0.43	0.88	0.94
RISK	0.53	0.26	0.41	0.84	0.83
PORT	0.50	0.42	0.46	0.93	0.76
XPORT	0.78	0.46	0.47	0.90	0.41
LAW	0.35	0.24	0.41	0.83	0.89
DP	0.87	0.14	0.35	0.63	0.93
DPM	0.80	0.24	0.38	0.72	0.79

Representative Table rows; darker cells are higher/better-oriented values

Fig. 3. Value-profile matrix for selected systems. Rows follow the labels in Table 3, and every cell is oriented so darker/larger values are better under the stated metric interpretation. This figure should be read as a profile comparison, not as a single ranking.

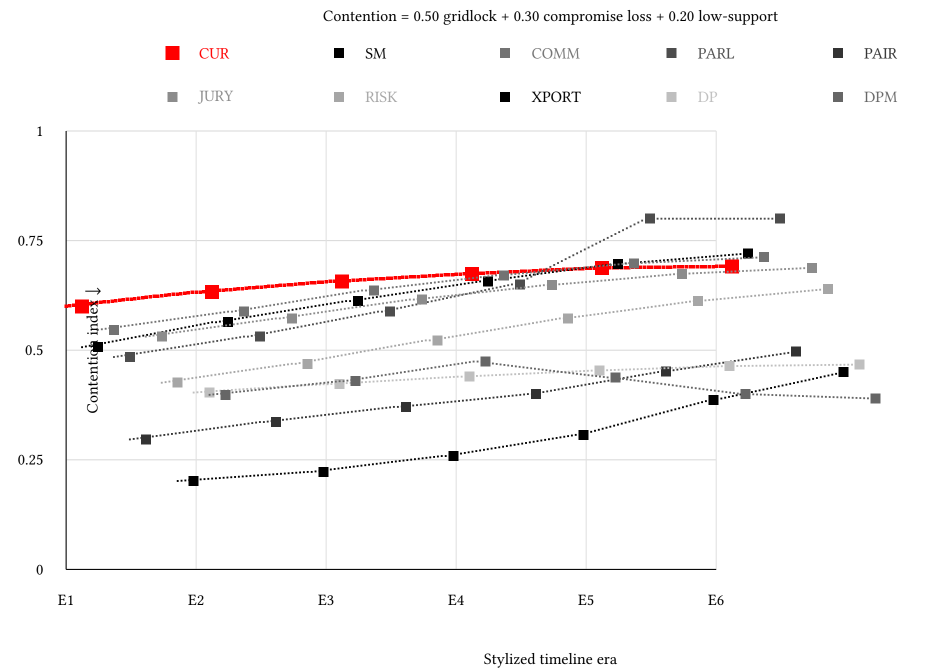


Fig. 4. Rising-contention stress path from the main comparison campaign. The index combines gridlock, compromise loss, and weak public-mandate passage. Lower is better. This is a parameterized stress path, not an endogenous causal model of polarization.

Three patterns are observed. First, thresholds alone are weak design language: supermajority and conventional baselines suppress weak mandates by suppressing throughput, while open default enactment reverses the problem. Second, agenda and content mechanisms matter across families: alternatives, amendment tournaments, petitions, open calendars, discharge, citizen review, initiatives, objection windows, and harm rules change who can force attention and at what cost. Third, reversibility is separate: judicial and law-registry review reduce exposure to weak or rights-risky laws but can increase volatility.

The exploratory result does not support a single dominant rule. The portfolio hybrid turns the pattern into a testable design: fast lanes for low-risk proposals, stronger consent or revision for risky proposals, proposer accountability, influence disclosure, alternatives before binary ratification, affected-group protection, and review of uncertain laws. Its mixed result is useful: combining good mechanisms improves several risk diagnostics, but adds administrative load and does not dominate simpler pairwise alternatives.

6 Limitations

The model is stylized. It uses one primary policy dimension, generated legislators, generated support and benefit, simplified issue domains, and abstract lobbying units. Public-mandate diagnostics distinguish salience and costly affected-group pressure from cheap amplified volume, but they are still proxies. Because public benefit is internal to the generator, welfare comparisons are conditional on model assumptions. Systems that screen extreme, uncertain, or lobby-heavy bills may look better partly because the generator often encodes those features as lower expected public value. Adversarial cases partially relax this assumption, but welfare remains a synthetic signal, not an empirical social-welfare estimate.

The calibration layer screens conventional baselines against empirical ranges, and the empirical bridge currently includes one raw Congress.gov bill-progression sample. That sample is useful for agenda-access plausibility, not sufficient for validation. Optional adapters also cover roll calls, topics, lobbying, district opinion, committees, campaign finance, court review, implementation, law revision, and comparative institutions; fixture samples exercise adapter shapes but are not validation data. Strategic behavior remains bounded: proposers and lobby groups adapt, and norm erosion can raise delay, but parties, committees, coalitions, courts, agencies, media, and elections do not yet co-evolve. Chamber scenarios are sensitivity prototypes. Citizen panels, initiatives, objection windows, tournaments, audits, challenge tokens, judicial review, and law-registry review are optimistic prototypes whose costs are represented only indirectly. Package bargaining models side payments, delay, jurisdictional trades, and distributive load, not legal text or appropriations.

7 Reproducibility

The submitted manuscript is anonymized for double-blind review. The reproducibility package contains source code, generated reports, the ODD+D appendix, and scripts. `make paper-checks` regenerates the campaign, diagnostics, figures, PDFs, word-count check, anonymity scan, figure-label and table consistency checks, and PDF manifest. `make supplement-anonymous` builds the double-blind supplement. Optional fixture samples can be refreshed with `make fetch-validation-samples`; they remain separate from raw validation inputs. `make build-bill-progression-raw` rebuilds the current Congress.gov bill-progression validation sample when a Congress.gov API key is available.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. Large language model tools were used for software implementation assistance and for drafting, editing, and reorganizing manuscript text. The author reviewed and is responsible for all code, experiments, reported results, text, and citations. No large language model is listed as an author.

8 Conclusion

This paper argues for computational mechanism search as a way to study legislative design. Default enactment is one stress case, but the broader contribution is comparing ordinary and unusual institutional bundles under common assumptions. The prototype suggests that compromise and productivity depend on combinations of agenda access, contestation, mediation, alternative selection, anti-capture safeguards, affected-group protection, and reversibility rather than any single passage threshold. The portfolio hybrids are synthesized candidates; the next step is to fit and stress-test their public-opinion, influence-system, court-review, and bargaining submodels rather than treating either as final.

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